

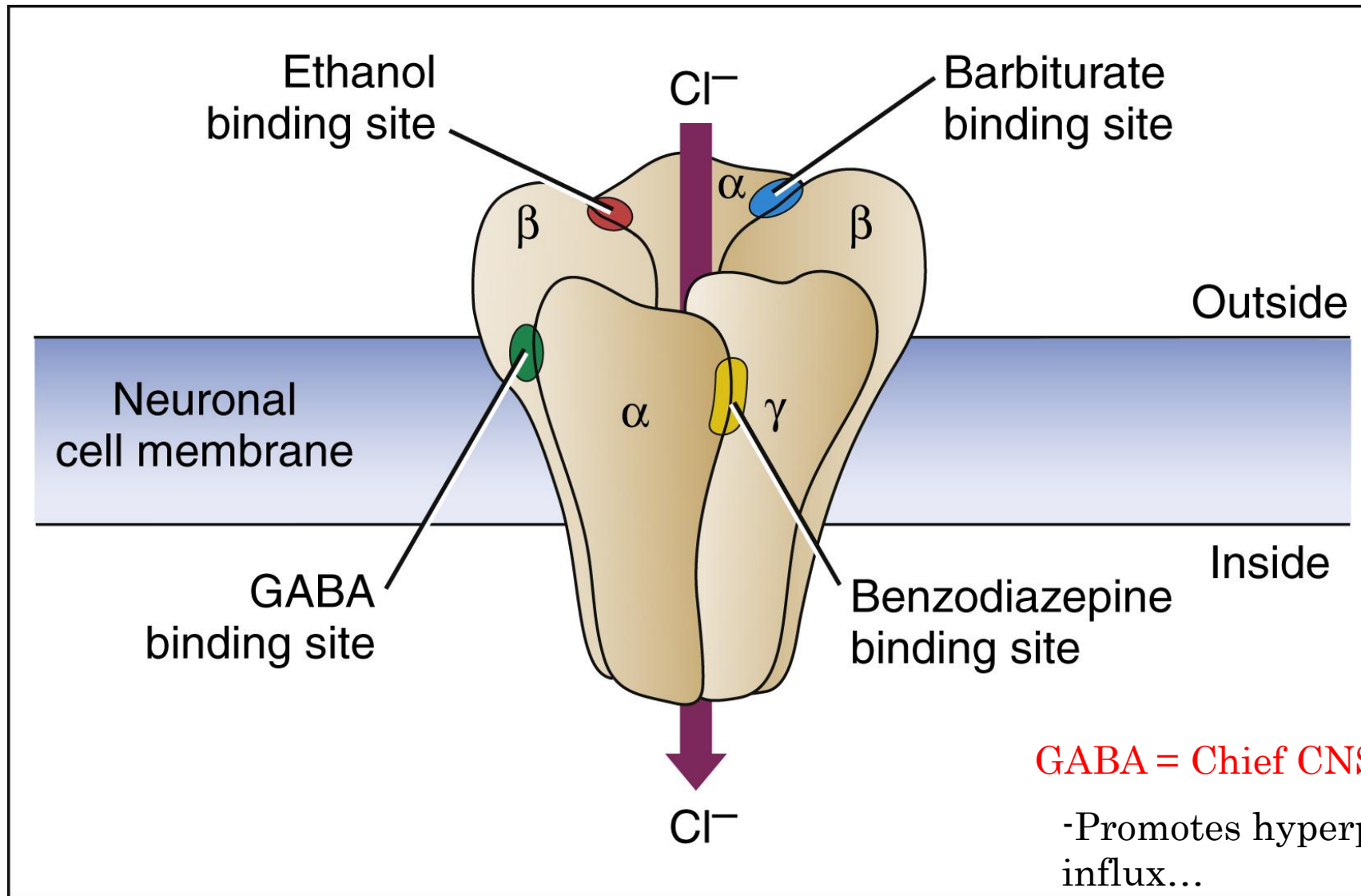
DRUGS ACTING ON NERVOUS SYSTEM

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ANXIOLYTIC AND HYPNOTIC DRUGS

- Anxiety is an unpleasant state of tension, apprehension, or uneasiness
- Anxiolytic drugs- relieve anxiety and they can trigger sleep (hypnosis)
- Drugs used for anxiety are called antianxiety agents or anxiolytics
- Drugs used for sleep known as hypnotics
- Anxiolytic-hypnotic drugs include
 - Benzodiazepines... ↑ the frequency of Cl^- channel opening of GABA_A rec
 - Barbiturates... ↑ duration of Cl^- channel opening of GABA_A rec.

Site of Action of Drugs on the GABA_A Complex



These drugs are called CNS depressants b/c they potentiate GABA effect

GABA = Chief CNS inhibitory neurotransmitter

-Promotes hyperpolarization via $\uparrow \text{Cl}^-$ influx...

i. Benzodiazepines(BZs)

- First choice for treating anxiety and insomnia as they have fewer side effects compared to barbiturates
- *Uses of Benzodiazepines*
 - **Alprazolam**- Anxiety, panic, phobias
 - **Diazepam**- Anxiety, sedation, seizure, muscle relaxation, withdrawal states
 - **Lorazepam**- Anxiety, preopsedation, status epilepticus (IV)
 - **Midazolam**- anesthesia IV and also for endoscopic procedure
 - **Temazepam**- Sleep disorders
- S/E- addictive ability(dependance) and tolerance ability
- Withdrawal signs of BZs:
 - Rebound insomnia
 - Anxiety, tremor etc

ii. Non-Benzodiazepines

- Non-BZ drugs: such as Zolpidem and zaleplon
- Binds to GABA
- Less tolerance and abuse ability than BZs
- Use- anxiety and for short-term management of insomnia
- BZs and non-BZs overdose can be treated by Flumazenil which blocks BZs GABA_A binding site (Flumazenil is BZs& non-BZs antagonist)

iii. Barbiturates

- Are general inducers of cytochrome P450s- they induce metabolism of oral contraceptives, carbamazepine, phenytoin, warfarin, etc.
- **Uses of barbiturates:**
 - Pentobarbital, Phenobarbital - used for seizures
 - Thiopental is used for induction of anesthesia
- S/E-CNS and Resp depression
- Withdrawal signs of barbiturates and ethanol:
 - Anxiety
 - Agitation
 - Life-threatening seizures
- Have more efficacy than BZs- it may lead to coma
- Are powerful respiratory depressants that can be fatal with over dosage.
- No Barbiturate antagonist

ANTIDEPRESSANTS

- Used primarily to relieve symptoms of depression.

Clinical Features:

- The principal symptoms
 - ✓ are depressed mood, loss of pleasure or interest, sadness and hopelessness, changes in sleep patterns and appetite, loss of energy, and suicidal thoughts.

Pathogenesis:

- Neurotrophic Hypothesis;
 - ✓ brain derived neurotrophic factor (BDNF)
- *monoamine hypothesis of the depression*;
 - ✓ depression is caused by a functional insufficiency of monoamine NTs (noradrenaline, serotonin, or both).

Antidepressant drugs



- **Depression**- likely due to Monoamine (serotonin) deficiency
- **Serotonin**- Involved in mood, depression and pain regulation
- Antidepressants-modify monoamine NT in the brain
 1. Selective serotonin reuptake inhibitors(SSRIs)
 2. Serotonin/norepinephrine reuptake inhibitors(SNRI)
 3. Tricyclic anti depressants (TCAs)
 4. Monoamine oxidase inhibitors(MAOI)...inhibit MAO-A enzyme

Treatment Modalities:

- Pharmacotherapy
- electroconvulsive therapy

1. Selective Serotonin Reuptake Inhibitors (SSRIs)

- specifically inhibit serotonin reuptake,
- the SSRIs have little blocking activity at muscarinic, α -adrenergic, and histaminic H₁ receptors
- Thus, no TCA S/E (hypotension, sedation, or anti-cholinergic effects like dry mouth or blurry vision); overdose doesn't cause cardiotoxicity
- drugs of choice in treating depression
- ***Mechanism of action***—Selective inhibition of serotonin reuptake to intensify transmission at serotonergic synapses; produces CNS excitation rather than sedation.
- ***Therapeutic uses***—For major depression, obsessive compulsive disorder
- **SSRIs:** fluoxetine, sertraline, paroxetine Citalopram
- S/E- loss of libido, weight gain, anxiety, headache

2. Serotonin/norepinephrine reuptake inhibitors(SNRI)

- *E.g Venlafaxine, desvenlafaxine, and duloxetine*
- inhibit the reuptake of both serotonin and norepinephrine
- may be effective in treating depression in patients in whom SSRIs are ineffective.
- The SNRIs, unlike the TCAs, have little activity at α -adrenergic, muscarinic, or histamine receptors and, thus, have fewer adverse effects

3. Tricyclic Antidepressants (TCAs)

- Amitriptyline, nortriptyline, imipramine, desipramine
- Third-choice drugs for depression
- ***Mechanism of Action***—Block monoamine reuptake like SNRIs
 - But also block α_1 , Muscarinic and H1 receptors
- ***Therapeutic Uses:***
 - Depression—Takes 1–3 weeks for initial responses
 - Bipolar disorder, chronic insomnia, attention-deficit/hyperactivity disorders
 - Other uses—Chronic insomnia and panic disorder
- ***Adverse Effects***—Orthostatic hypotension is most common; anticholinergic effects; diaphoresis; sedation; cardiac toxicity is most serious

4. Monoamine oxidase inhibitors (MAOIs)

- MAO enzymes inactivate any excess neurotransmitters (for example, norepinephrine, dopamine, and serotonin)
 - MAO-A(nonselective) is primarily responsible for NE, 5-HT & tyramine metabolism
 - MAO-B(selective) is more selective for Dopamine metabolism
- Indicated for depressed patients who are unresponsive or allergic to TCAs and SSRIs
- **S/E-** hypertensive crisis (occipital headache, stiff neck, tachycardia, nausea, hypertension, cardiac arrhythmias, seizures)
- **Other S/E-** drowsiness, orthostatic hypotension, blurred vision, dry mouth

ANTI PSYCHOTIC DRUGS

Schizophrenia

- Geek words: *Skhizo*(to split) & *Phern* (mind) w/c means the split b/n the emotions & the intellect
- It is one particular kind of psychosis

Main clinical features

Positive symptoms : delusions, hallucinations & thought disorder

Commonly occur in acute phase of the illness

Usually respond to antipsychotic drug therapy

Negative symptoms : apathy, social withdrawal & lack of drive

Occur in the chronic pahse of illness

Tend to be resistant to dug therapy

Schizophrenia

- It occurs in about 1% of the population and is a chronic and disabling disorder
- Schizophrenia has a strong genetic component
- The antipsychotic drugs are divided into first- and second-generation agents.
- **I. First generation antipsychotic** - block dopamine D2 receptors.
 - *E.g Chlorpromazine, Fluphenazine, Haloperidol*
- **II. Second-generation antipsychotic** drugs (also called “atypical” antipsychotics)
 - block both serotonin and dopamine receptors
 - *E.g Clozapine, Olanzapine, Risperidone*
- S/E = Parkinson like symptoms of bradykinesia, rigidity, and tremor
=Anticholinergic and antihistamine effect

ANTIEPILEPTIC AGENTS

- Seizures result from episodic electrical discharges in cerebral neurons associated with prolonged depolarization, during which sustained, high-frequency, repetitive firing (SHFRF) occurs, followed by prolonged hyperpolarization. The goal of drug management is restoration of normal patterns of electrical activity.
- Mechanism of seizure
 - depolarization of ions (increase Na^+)
 - depolarization of ions (increase Ca^{++} influx)
 - By decreasing GABA effect (less CNS inhibition i.e increase excitation)
....(By decreasing hyperpolarization...decrease Cl^- influx)
 - By increasing Glutamate effect (increase CNS excitation)

Anti-epileptics

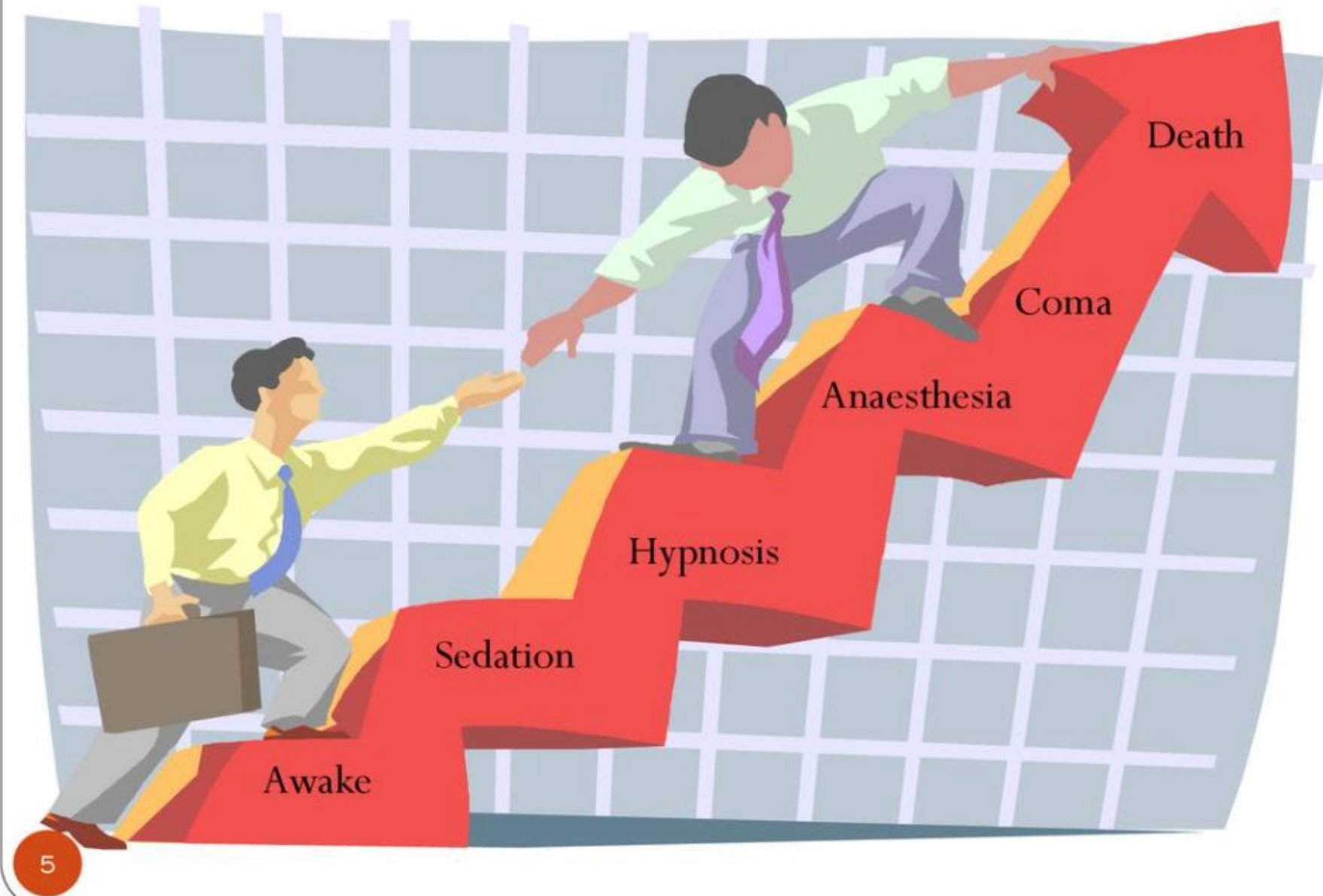
- Mechanisms of action:
 - ↓ axonal conduction by preventing Na^+ influx through fast Na channels—carbamazepine, phenytoin
 - ↑ inhibitory tone by facilitation of GABA-mediated hyperpolarization—barbiturates, benzodiazepines
 - ↓ excitatory effects of glutamic acid—lamotrigine, topiramate (blocks AMPA receptors); felbamate (blocks NMDA receptors)
 - ↓ presynaptic Ca^{2+} influx through type-T channels in thalamic neurons—ethosuximide and valproic acid

ANESTHETICS

- **Anesthesia** is a state of temporary loss of sensation. It may include some or all of analgesia, paralysis, amnesia, and unconsciousness(GA)
- **General Anesthetics(GA)**
 - Used as adjunct to surgical procedures in order to render the patient unaware of and unresponsive to painful stimulation
 - Anesthetic potency is closely correlated with lipid solubility
 - Types of general anesthetics
 - *On the basis of route of administration*

18th Century Surgery





General anesthetics (GA)

- Physiologic state induced by GA
 - analgesia
 - amnesia
 - loss of consciousness
 - inhibition of sensory & autonomic reflexes
 - skeletal muscle relaxation

Phases of GA

- Induction
- Maintenance
- Recovery

i. Inhalational Anesthetics

- Are gases or volatile liquids
 - Desflurane
 - Sevoflurane
 - Isoflurane
- Enflurane
- Halothane
- Ethoxyflurane
- Nitrous oxide
- Diethyl ether
- Metabolism & Toxicity
 - Chloroform: hepatotoxicity associated with free radical formation
 - Enflurane & sevoflurane: generate fluoride, but lower level
 - Halothane: 30% metabolized [liver toxicity]

- **Halothane**
 - Widely used, but now declining in favor of **isoflurane**
 - Nonexplosive & nonirritant
 - Highly potent & easily produce respiratory & cardiovascular failure
 - In normal anesthetic concentration, it causes fall in BP.
- Adverse effect of halothane
 - Halothane sensitizes the heart to adrenaline, & tends to cause **cardiac dysrhythmias**
 - **Hepatotoxicity** (Trifluoroacetic acid reacts covalently with proteins)
 - **Malignant hyperthermia** ($\uparrow \text{Ca}^{2+}$ from sarcoplasmic reticulum $\rightarrow$ metabolic heat production in skeletal muscle)

- **Nitrous oxide (N₂O)**
 - Low potency, must be combined with other agents.
 - **Rapid induction** and recovery
 - Good analgesic properties
 - During recovery, produce a transient hypoxia
- **Enflurane**
 - Less metabolism than halothane; **less risk of toxicity.**
 - Faster induction & recovery than halothane [accumulation in fat)
 - **Risk of seizures**
 - Can induce **malignant hyperthermia**

ii. Intravenous anesthetics

- Do not require specialized vaporizer equipment.
- Commonly used for induction of anesthetic
- **Benzodiazepine**: sedation and amnesia
 - Diazepam, lorazepam, & **midazolam**(most commonly)
 - Premedication: sedation & amnesia
- **Short acting barbiturates**
 - Thiopental: commonly used for induction of anesthesia
- **Opioid analgesics**
 - **Fentanyl & droperidol** together produce analgesia and amnesia
 - **Fentanyl** contributes cardiovascular stability, enhanced sedation and profound analgesia
- **Propofol**
 - May give profound hypotension
 - Postoperative nausea and vomiting is less common
 - has replaced *thiopental* as the first choice for induction of general anesthesia
- **Ketamine**
 - S/E-Produces dissociative anesthesia [amnesia and analgesia, with or without actual loss of consciousness.

Local anesthetics(LA)

- **Local anesthesia**- Loss of sensation on area around injection
- Usually accompanied by epinephrine

Metabolism and excretion

- Converted in the liver or plasma to more water-soluble metabolite
- *i. Ester type* local anesthetics are hydrolyzed very rapidly in the blood by butyryl cholinesterase (pseudocholinesterase)
 - E.g Cocaine, procaine, tetracaine
- *ii. Amide types* hydrolyzed by liver microsomal enzyme
 - E.g lidocaine, bupivacaine, mepivacaine
- **Pharmacodynamics of LA**
- → Blockade of Na⁺ channel
- Use of LA
 - To take a skin biopsy or breast biopsy,
 - stitching a deep cut etc

- **ANALGESICS**

- Drugs that selectively relieve pain with out significant change on patient consciousness
- Act on CNS or periphery pain mechanism
- **Classes:**
- Opioid analgesics – for severe pain
 - Agonist
 - Antagonist
- Non-opioid analgesic – for mild pain
 - NSAIDs like Aspirin, Ibuprofen, diclofenac

Pathophysiology of pain

Noxious stimulus



NSAIDs



Release of inflammatory substances
(PG, His, 5-HT, Bks, Sub.P)



Transduction (generation & electrical impulses)



Transmission (conduction by nerve fibers)

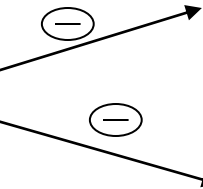


Modulation (Modification with spinal cord)



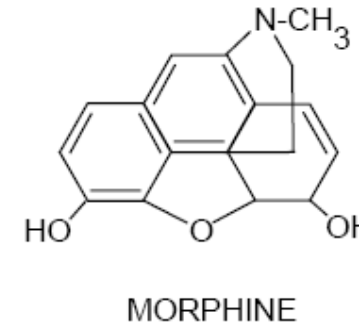
Perception

Opioids



Opioids

- Derivative of opium poppy
- Prototype: morphine
- Naturally occurring-morphine, codeine
- synthetics; pethidine, meperidine, fentanyl, tramadol
- Act on endorphin receptors:
 - Mu (most important)
 - Kappa
 - Delta
 - All belong to GPCR family
 - Opioid receptors are in the CNS, GI tract and the urinary bladder.
- Actions of Opioids
 - Analgesia
 - Euphoria/Dysphoria



Opioid Antagonist

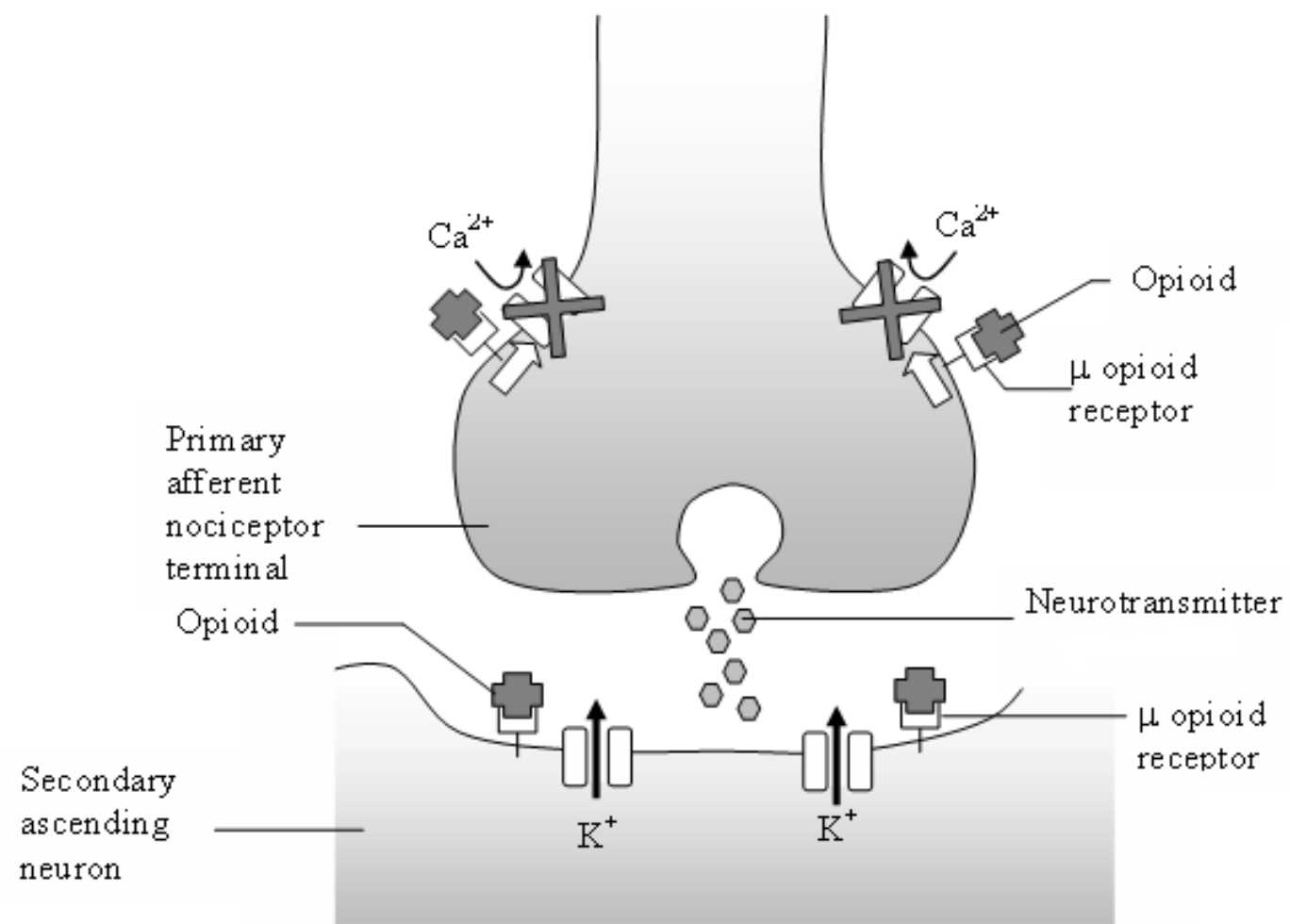
- naloxone
- naltrexone

Opiate receptor interactions

Drugs	Receptor		
	mu	kappa	delta
<u>Full agonists</u>			
Morphine	+++	++	+
Fentanyl	++++	+	
Hydromorphone	+++	++	
Methadone	+++	++	
Meperidine	++	++	
Codeine	+		

Mechanism of action of opiates

- Opioids produce analgesia by binding to specific GPCRs located in brain & spinal cord regions involved in the transmission & modulation of pain
- Inhibit adenylyl cyclase $\rightarrow$ $\downarrow$ cAMP
- Reduce neuronal excitability
 - b/c of the $\uparrow$ K^+ conductance causes hyperpolarization of the membrane
- Reduce transmitter release
 - Due to inhibition of Ca^{2+} entry



Adverse reaction

- severe respiratory depression
- Hypotension
- Sedation
- Miosis
- Constipation
- Urinary retention
- Addiction
- death from acute opioid overdose can occur.

Non Steroidal Anti inflammatory Drugs(NSAIDs)

- **Mechanism of action**

- Inhibits cyclooxygenase(COX) enzyme & hence prostaglandin synthesis
- COX-1 is a primarily constitutive in most normal cells & tissues
- COX-2 is induced isoform (in inflammation)

- **Clinical use**

- All NSAIDs have
 - Anti-inflammatory
 - Antipyretic
 - Analgesic effects

However, acetaminophen (paracetamol) is devoid of anti-inflammatory activity

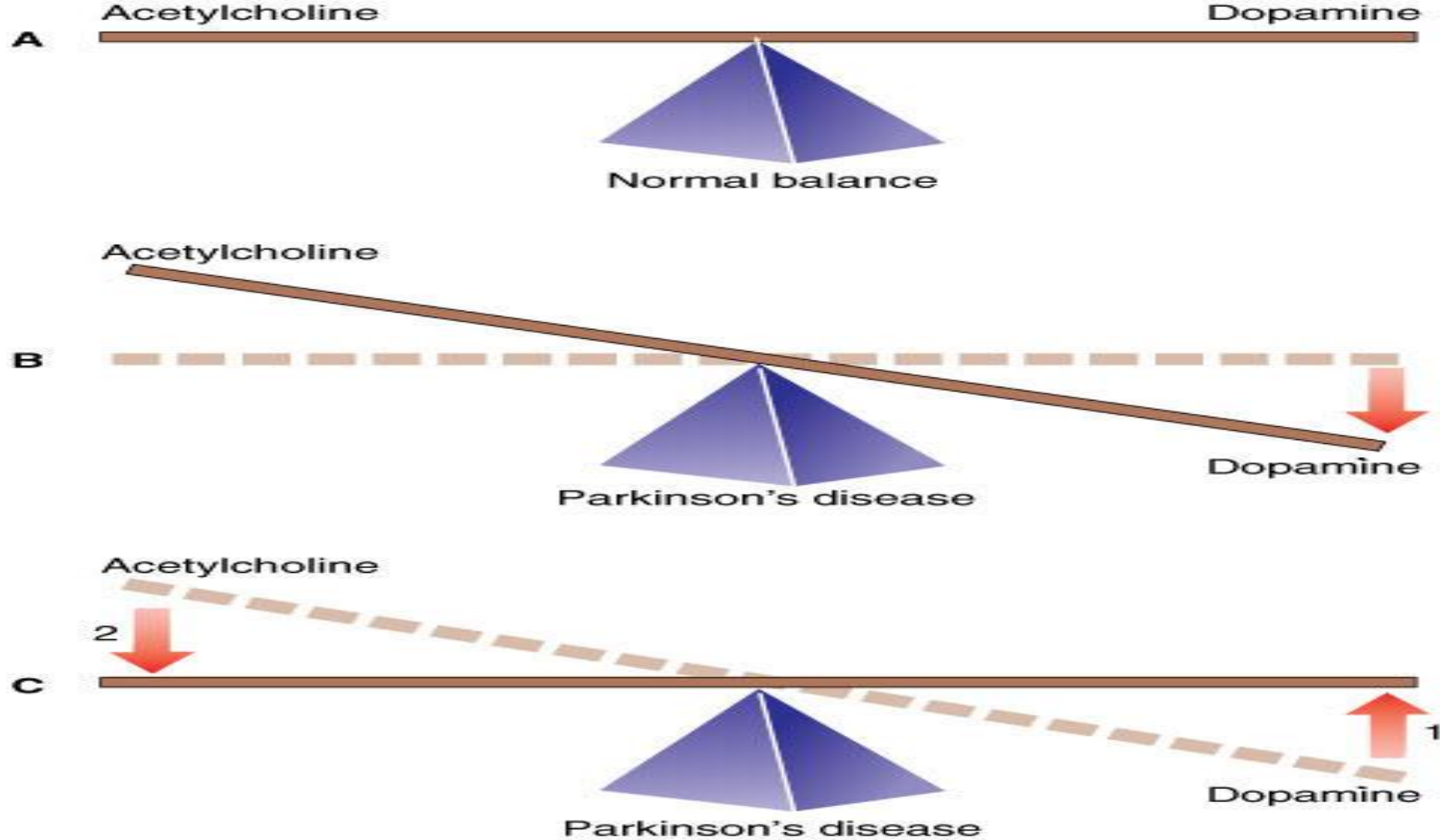
Neurodegenerative diseases

Most common neurodegenerative conditions

- Parkinson's disease (PD)
- Alzheimer's disease (AD)
 - The disorders are characterized by progressive & irreversible loss of neurons from specific regions of the brain

ANTIPARKINSONIAN DRUGS

- Parkinsonism is a progressive neurological disorder of muscle movement,
- The **cause** of Parkinson's disease
 - is idiopathic,
 - *D₂ antagonists* may cause reversible parkinsonism
- **Pathogenesis** -the disease is associated with destruction of dopaminergic neurons in the substantia nigra (parts of the basal ganglia system that are involved in motor control)
 - **Dopamine**- is involved in movement, attention, learning, motivation, reward
 - Low Dopamine
 - High Ach
 - *N.B-Overactive Dopamine- Schizophrenia (Rx- Antipsychotics)*

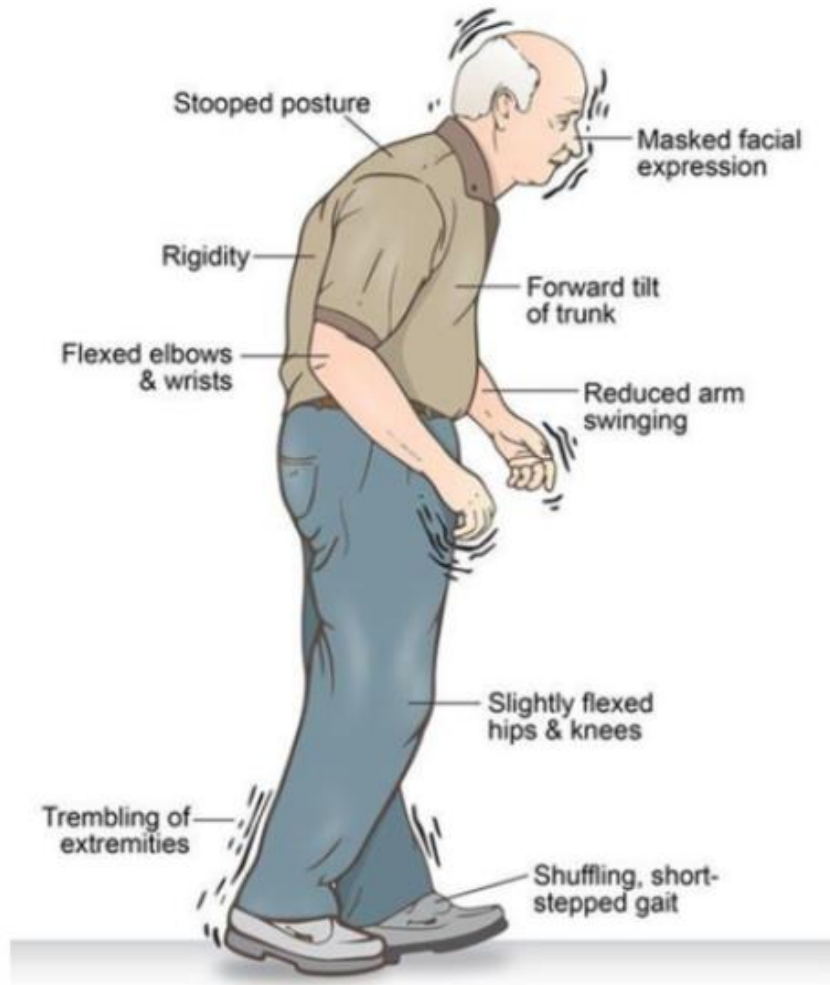


- A,** Normal balance of acetylcholine and dopamine in the CNS.
B, In Parkinson's disease, a decrease in dopamine results in an imbalance.
C, Drug therapy in Parkinson's disease is aimed at correcting the imbalance between acetylcholine and dopamine. This can be accomplished by either
 1. increasing the supply of dopamine or
 2. blocking or lowering acetylcholine levels.

Fig. 14-1. The neurotransmitter abnormality of Parkinson's disease.

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- The basal ganglia are responsible for controlling automatic movements of the body by the action of dopamine (DA)
- Degeneration of basal ganglia in the deeper grey matter of the brain (**substantia nigra**) causes **PD**
- In PD, there is extensive destruction of the dopaminergic neurons of the substantia nigra
 - DA deficiency
- **PD symptoms**- are tremor, muscular stiffness, rigidity, difficulty of walking, balance, and coordination and gait abnormalities.

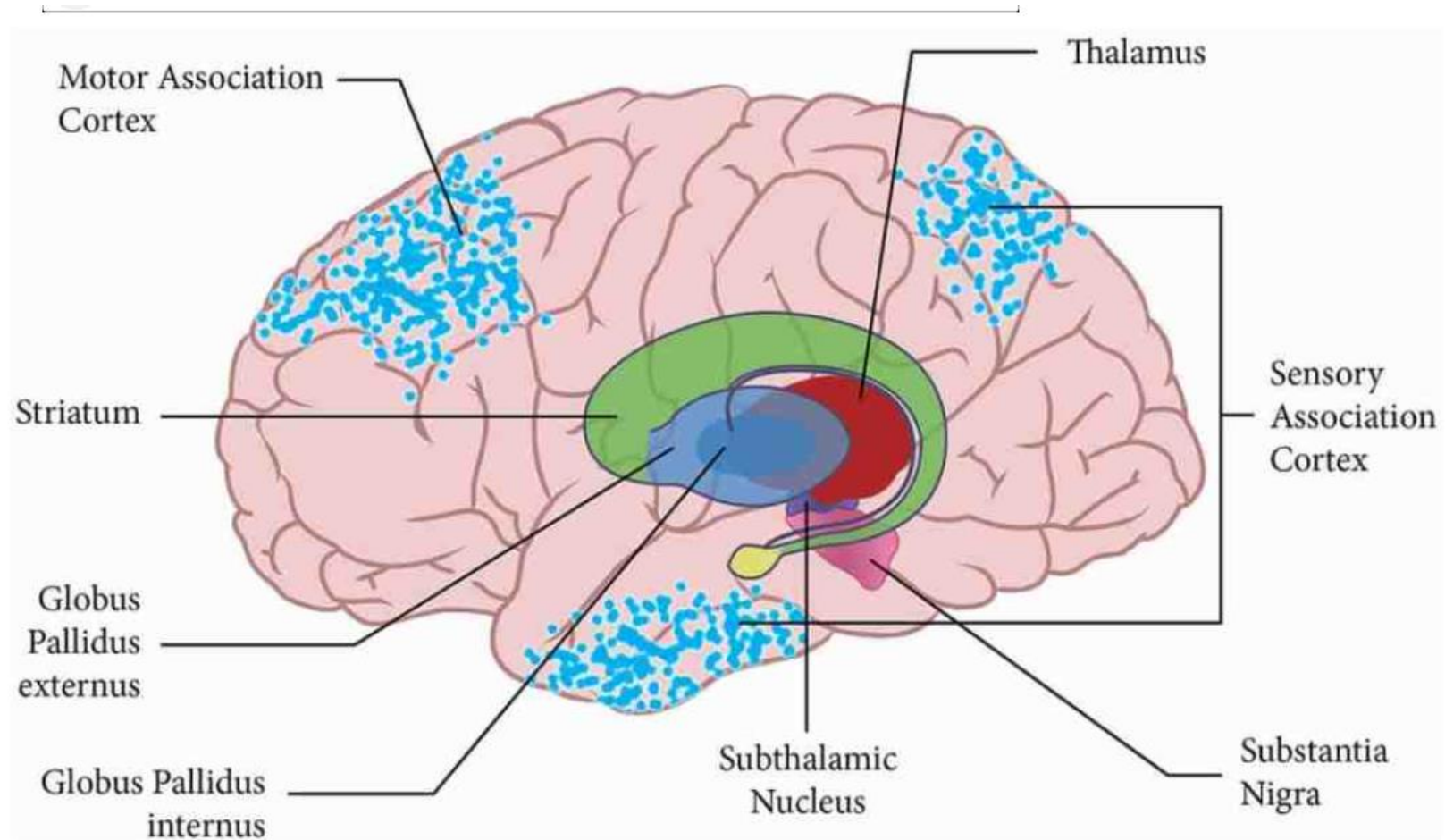


T Tremor: shaking, usually starting on one side

R Rigidity: stiffness of the limbs, neck, or trunk

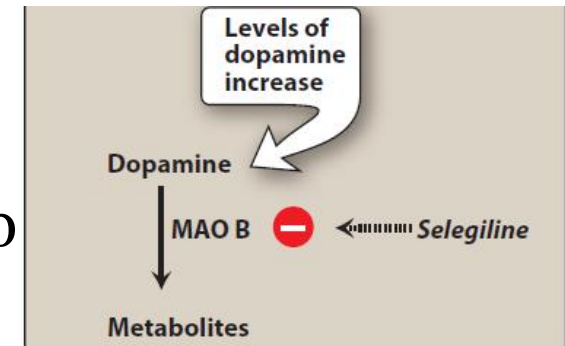
A Akinesia: loss or impairment in power of voluntary movement

P Posture and balance

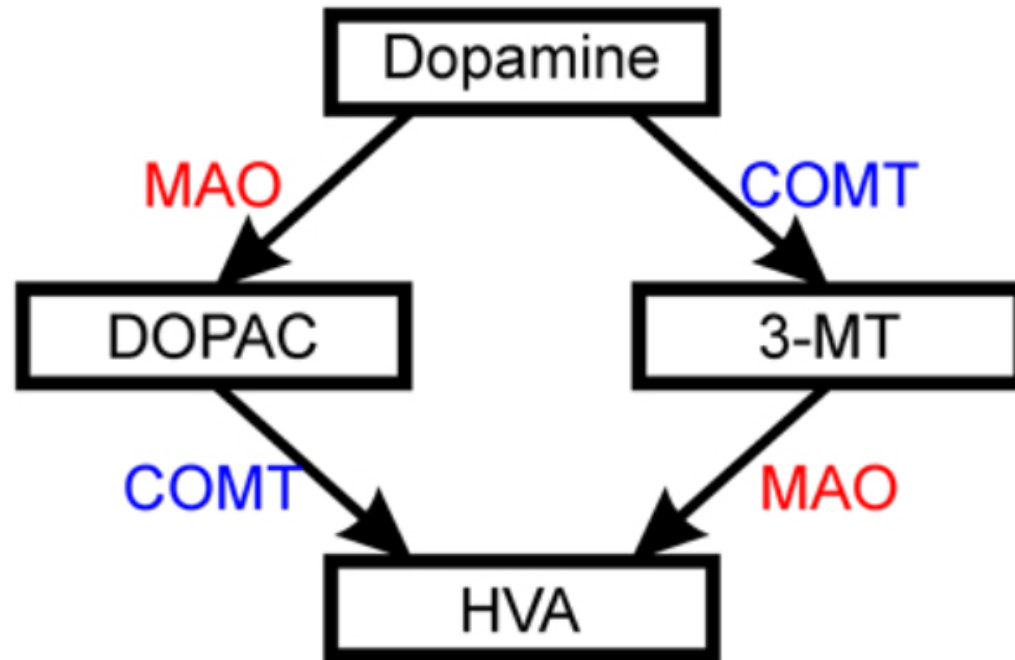


Treatment of Parkinson's disease

- Drugs which increase dopamine, act as dop or stimulate DR
- Drugs which decrease the effect of Ach
- 1. **Levodopa + Carbidopa**
- 2. Amantadine - ↑synthesis, blocks reuptake of dopamine from the surviving neurons
- 3. Dopamine receptor agonist- Bromocriptine, Pramipexole, Ropinirole
- 4. MAO-B Inhibitor- selegiline
- 5. Catechol-O-Methyltransferase (COMT) inhib
 - ✓ E.g Tolcapone & entacapone
- 6. Antimuscarinic agents – e.g. Benztropine, trihexphenidyl, Biperidine.



Dopamine metabolism



- **DOPAC** (3,4-dihydroxyphenylacetic acid)
- **3-MT** (3-methoxytyramine)
- **HVA** (homovanillic acid)
- **MAO** (monoamine oxidase)
- **COMT** (catechol-O-methyltransferase)

ALZHEIMER DISEASE

- Pathogenesis
 - Loss of cortical neurons, particularly cholinergic neurons.
- Due to
 - accumulation of senile plaques (β -amyloid accumulations), formation of numerous neurofibrillary tangles,
 - Excitotoxic actions resulting from overstimulation of NMDA-glutamate receptors
- Current therapies aim to either improve cholinergic transmission within the CNS

Treatment of Alzheimer Disease

1. Acetylcholinesterase inhibitors

- The reversible AChE inhibitors approved for the treatment of mild to moderate Alzheimer's
 - *donepezil*,
 - *galantamine*, and
 - *rivastigmine*.
- All of them have some selectivity for AChE in the CNS, as compared to the periphery

2. NMDA receptor antagonist eg *memantine*

- glutamate binds with NMDA receptors in the CNS